

COMP4616 – Fundamentals of Machine Learning

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PERFORMANCE METRICS

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

Performance Metrics

Confusion Matrix

- Confusion matrix is a table that summarizes the performance of a classification model.
- It helps us understand how well the model's predictions align with the actual outcomes.
- The confusion matrix has four components:
 - **True Positives (TP)**: Examples correctly predicted as positive.
 - **True Negatives (TN)**: Examples correctly predicted as negative.
 - **False Positives (FP)**: Examples incorrectly predicted as positive.
 - **False Negatives (FN)**: Examples incorrectly predicted as negative.

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

Confusion Matrix

- Example: Imagine classifying 100 people as pregnant or not pregnant with a test. We have:
 - 40 pregnant women (actual positive cases).
 - 60 non-pregnant individuals (actual negative cases).
- However, we occasionally get false positives and false negatives with this test as shown in the confusion matrix below:
- So to understand how our test performs, we will use the confusion matrix along with performance metrics.

	Predicted Positive	Predictive Negative
Actual Positive	TP = 30	FN = 10
Actual Negative	FP = 5	TN = 55

Accuracy

- Accuracy measures the proportion of correct predictions (both true positives and true negatives) out of the total number of predictions made .
- It is calculated with the formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Accuracy of our test would be then

$$Accuracy = \frac{30 + 55}{100} = 0.85$$

	Predicted Positive	Predictive Negative
Actual Positive	TP = 30	FN = 10
Actual Negative	FP = 5	TN = 55

Accuracy

- Accuracy can be misleading in cases where the class distribution is imbalanced.
- It is easy to cheat with accuracy when we have a biased dataset.
- For example, in a dataset where 95% of the instances belong to one class (negative) and 5% to another class (positive), a model that always predicts the majority class (negative) would achieve 95% accuracy, despite failing to correctly identify any of the positive instances.
- Even if your accuracy is very high, your model did not learn to correctly predict the label/class of your positive examples.

Precision

- **Precision** (positive predictive value) tells us how often the model is correct when it predicts positive. It measures the accuracy of the positive predictions made by the model.
- Solves the cheating issue in accuracy metric with the biased datasets.
- It is calculated with the formula:

$$Precision = \frac{TP}{TP + FP}$$

- For example, in a spam detection system:
 - **True Positives (TP)**: Emails correctly identified as spam.
 - **False Positives (FP)**: Legitimate emails incorrectly identified as spam.
- High precision indicates that the model is effective in making accurate positive predictions, though it may also miss some actual positives (false negatives).
- Therefore, **precision** is often considered together with **recall** to provide a balanced evaluation using metrics like the **F1 score**, which is the harmonic mean of precision and recall.

Recall

- **Recall (sensitivity or true positive rate)** is the ratio of correctly predicted positive observations to all actual positive observations.

$$Recall = \frac{TP}{TP + FN}$$

- **Recall** is important in scenarios where it is crucial **to identify as many positive instances as possible**, such as in medical diagnosis or fraud detection, where missing a positive instance can have serious consequences.
- For example, in a medical test for a disease:
 - **True Positives (TP)**: Patients correctly identified as having the disease.
 - **False Negatives (FN)**: Patients who have the disease but were not identified by the test.
- High recall indicates that the model is effective in identifying positive cases, though it may also capture more false positives (negatives identified as positives), which would affect precision.
- Thus, recall is often considered together with precision to provide a balanced evaluation using metrics like the F1 score.

F1 score

- The **F1 score** is a metric used to evaluate the performance of a classification model by combining both **precision** and **recall** into a single measure.
- It is particularly useful when you need to balance the trade-off between precision and recall.
- The **F1 score** is the **harmonic mean of precision and recall**, giving equal weight to both metrics.
- The F1 score ranges from 0 to 1, where 1 indicates perfect precision and recall, and 0 indicates the worst performance.
- It is especially useful in cases where there is an **uneven class distribution**, and **one class is much more frequent than the other**.

$$F1\ score = 2 \times \left(\frac{Precision \times Recall}{Precision + Recall} \right)$$

F1 score

- For example, in a medical test for a rare disease:
 - **High precision** means most of the patients identified as having the disease actually have it.
 - **High recall** means the test successfully identifies most patients who have the disease.
 - The **F1 score** provides a single measure that balances these two aspects, indicating the overall effectiveness of the test in correctly identifying patients with the disease.